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Microfluidic Evaluation of Hybrid Low Salinity Waterflood EOR in Carbonate Reservoirs

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Abstract

One of dominant reactions in low salinity waterflood (LSW) enhanced oil recovery (EOR) mechanism is a fluid-fluid interaction (FFI) that forms micro-dispersion (MD) at the interface of oil and LSW in contact. The MD ratio (MDR), defined as an increase of water content in MD phase compared with that of original water content in oil, is considered to be linearly associated with an oil recovery improvement in tertiary mode LSW core floods. For further understanding of the role of MD formation, we applied microfluidics technology to visualize displacement process with the FFI mechanism to increase oil recovery. Referred to a series of 2D CT-image stack scanned from the target carbonate core pieces, a representative microfluidic chip was manufactured with reproducing typical pore throat paths and carbonate-corresponding wettability. The reference pieces were collected from a vicinity of locations where plug-cores were taken for core floods so that the microfluidic tests could use sister core information for a fair comparison purpose with the previous core flood study. A total of four microfluidic flood experiments were performed evaluating LSW injection (diluted to 1% sea water (SW), TDS 430 ppm) or hybrid LSW (1% SW containing 2wt% diethyl ketone) injection compared to SW injection (TDS 43,000 ppm).

In the past tertiary mode core flood tests, the highest MDR oil, STO-L2, showed noticeable oil recovery increment (+11% IOIP) with hybrid LSW while a clear increment (+3% IOIP) with pure LSW. Thus, two series of microfluidic tests were performed. The first series of runs (i.e., continuous secondary mode injection of SW vs LSW) evaluated pure LSW injection using a synthetic pattern shape of porous media that represented relatively larger spaces than the actual ones so that FFI reaction can be promoted due to more contact opportunity between LSW and oil. The tests captured clear snapshots of emulsification with increasing trend of differential pressure during the LSW injection stage. The second series of runs evaluated hybrid LSW injection (i.e., after SW injection, tertiary mode injection of hybrid LSW or followed by chase SW) using the realistic microfluidic chip that has smaller pore throat than the first chip. To secure sufficient MD forming reaction time, a 7-day ageing period was set during the hybrid LSW injection stage. Consequently, the post-ageing flood showed a clear increase in differential pressure with additional oil

recovery (+8.4% IOIP) while the pre-ageing stage reached plateau oil recovery. The similar oil recovery increment (+8.6% IOIP) was confirmed by switching from hybrid LSW to chase SW injection. During both pure and hybrid LSW injection stages, we observed a similar behavior that increased differential pressure. Those common trends imply varying viscosity and/or interfacial viscoelasticity of LSW/hybrid LSW contacted oils.

Introduction

Many studies showed a potential of low salinity waterflood (LSW) enhanced oil recovery (EOR) in carbonates (Yousef et al. 2010; Austad et al. 2012; Zahid et al. 2012; Mahani et al. 2015, 2017; Sohrabi et al. 2017; AlHammadi et al. 2018; Masalmeh et al. 2019; Mehraban et al. 2020; Alhammadj et al. 2021; Adila et al. 2022; Souayeh et al. 2024a; Tong et al. 2024). Based on much attention to LSW EOR in carbonate, many types of combination with other methods have been extended such as LSW with surfactants (LSS) (Veiskarami et al. 2020; Souayeh et al. 2021; Belhaj et al. 2022, 2023, 2024; Fakir et al. 2023; Kalam et al. 2023; Souayeh et al. 2024b) and LSW with polymers (LSP) (Al-Murayri et al. 2019; Song et al. 2022; Hassan et al. 2024). The synergized concept of LSW and chemical agents is to assist the role of chemical function by LSW. In recent studies, as a reverse concept of chemical assisting the role of LSW function, another type of hybrid LSW EOR with a combination of diethyl ketones (DEK) or dimethyl ketones (DMK) has been reported to increase oil recovery beyond pure LSW EOR increment (Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024b). In fact, under the continuous sequence of secondary sea water (SW) injection (TDS 43K ppm) followed by tertiary pure LSW or hybrid LSW injection (TDS 430 ppm), the tertiary stages achieved additional oil recovery of +3-11 % IOIP. However, the profiles showed slower oil production in the tertiary stages. For example, the hybrid LSW oil recovery started at 4 pore volume (PV) injection after switched from secondary to tertiary mode. This phenomenon motivated us to investigate the oil recovery behavior and/or mechanism. The LSW-reactive oil screened and used in the core floods was confirmed to form a noticeable micro-dispersion (MD) phase at the interface of oil and water phases. In the screening process, water contents in the MD phase were measured and interpreted to MD ratio (MDR) which is a dimensionless ratio based on bond water content in original oil. Because this is a key parameter that has a linear correlation with oil recovery increment in core floods, we consider that the main driving factor in oil recovery relies on the MD formation mechanism. Therefore, a common analysis such as interfacial tension and viscosity measurement was applied for elucidating the mechanism, but it could not distinguish any specific variation between MD phases and original crude oil (Yonebayashi et al. 2024b). The results urged us to consider alternative approach for understanding the mechanism.

The E&P industry has applied microfluidics technology to investigate/visualize how microscopically trapped oil in porous media can be recovered. A microfluidics system consists of a chip, microscopy, and flood controlling devices with temperature/pressure control system. There are many types of chip device such as systematic and/or random geometric shapes (Armstrong and Wildenschild. 2012; Tahir et al. 2020) and recently actual pore network shapes (Gaol et al. 2020; Tetteh et al. 2020). According to purposes, multi pattern shapes, for example connecting large and small permeability/porosity networks were adopted for evaluating two-phase fluid flow and/or mobility control method (Wolf et al. 2022; Afsharpour et al. 2024). The microfluidic method has been already adopted to many evaluation studies of not only LSW EOR (Saadat et al. 2020; Tetteh et al. 2020; Siadatifar et al. 2021; Mahmoudzadeh et al. 2022; Duits et al. 2023) but also other EOR options (Armstrong and Wildenschild. 2012; Quennouze et al. 2014; Yun et al. 2020; Ismaili et al. 2023). Microfluidic technology has been a useful tool to better understand how the physicochemical properties of each factor such as oil, water, pore network properties are linked.

According to our previous findings of MD formation attributed to interactions between LSW-reactive oil and brine, the visualizing of flooding behavior was considered essential for understanding oil recovery behavior and underlying mechanisms. Therefore, this study adopted the latest microfluidic technology. The

goal in this approach was to gain better knowledge of how much and how fast (or slow) remaining oil could be produced from synthetic reservoir pores reproducing a typical initial wetting condition of carbonates. The new knowledge was linked to the findings accumulated from the previous studies to discuss comprehensive understanding for further oil recovery.

Materials, Equipment, and Procedures

The target oil field is an offshore giant carbonate with two main reservoirs that are defined finely as multiple sub-reservoirs in the Middle East. The main reservoirs are defined as Upper (U) and Lower (L). Three sub-reservoirs in the L reservoir are defined as Upper (U), Middle (M), and Lower (L). Therefore, each sub-reservoir is named L(U), L(M), and L(L), respectively. The field commenced oil production in 1967 followed by pressure maintaining operations by dump-flood water, peripheral sea water injection, and crestal gas injection. In addition to those pressure maintenance deployment, EOR options have been studied for further oil recovery by reservoirs.

Fluid Properties (Oil and Water)

Among a total of four stock tank oil (STO) samples collected from all the sub-reservoirs, STO-L2 originated from an oil producer in the L(M) sub-reservoir was screened as the best LSW-reactive oil through fluid-fluid interaction (FFI) test. All the STOs were categorized as light oils with 40°API gravity. Synthetic brine was prepared by reproducing injection SW that has been currently used in the powered water injection scheme (conditioned as specific gravity 1.032 at 20 °C; pH 8.1). The LSW was made up by diluting SW to 1%. The brine properties are summarized in Table 1.

Table 1—Summary of Brine Compositions.

		Sea Water (SW)	Low Salinity Water (LSW) Diluted to 1% SW
Cations			
Sodium (Na ⁺)	mg/L	13,790	138
Calcium (Ca ²⁺)	mg/L	484	5
Magnesium (Mg ²⁺)	mg/L	1,590	16
Barium (Ba ²⁺)	mg/L	<1.0	<1.0
Potassium (K ⁺)	mg/L	470	5
Strontium (Sr ²⁺)	mg/L	8.6	0.09
Anions			
Chloride (Cl ⁻)	mg/L	23,260	233
Bicarbonate (HCO ₃ ⁻)	mg/L	154	2
Sulphate (SO ₄ ²⁻)	mg/L	3,230	32
TDS	mg/L	42,987	430

Fluid-Fluid Interaction Procedure

LSW-reactive oils form an MD phase at the interface of oil and brine under static condition while non-reactive oils stay with a clear oil/water interface. Those two types of oil are defined as positive and negative oil, respectively. The FFI test was established by industry to provide a simple but representative screening method (Masalmeh et al. 2019). The method can distinguish positive/negative oil based on MDR that means how water content increases in MD phase from that of original oil phase. The MDR is a practical key performance indicator to display potential of high LSW EOR performance. The most usefulness of MDR is a clear relationship with oil recovery increment in subsequent core flood evaluation under dynamic conditions. Namely, a positive oil that is favored to form MD may promise additional oil recovery by LSW EOR core floods. The FFI test procedure is described below (from step-1 to step-4) and summarized in Figure 1(a). Prior to the FFI test, candidate oil samples were pre-treated by filtration at 5 µm and centrifugation at 1400

rpm for 60 minutes to remove free water and solid contamination. The water content in the pre-treated STO was defined as bond water that was used to calculate MDR.

(Step-1) a separating funnel to be filled with 230 mL brine, followed by 30 mL crude oil to be added slowly with a pipette to avoid any physical disturbance to the crude oil phase,

(Step-2) the funnel to be stored in an oven at 140 °F (60 °C) for three days,

(Step-3) aqueous phase to be slowly depleted from the bottom while ensuring not to physically disturb the oil-water interface

(Step-4) outlet line of the funnel to be cleaned with toluene and methanol, and then crude oil to be sub-sampled (10-15 mL) from the interfacial area and stored in glass vial

(Step-5) water content of the sub-samples to be analyzed by Karl Fischer titration.

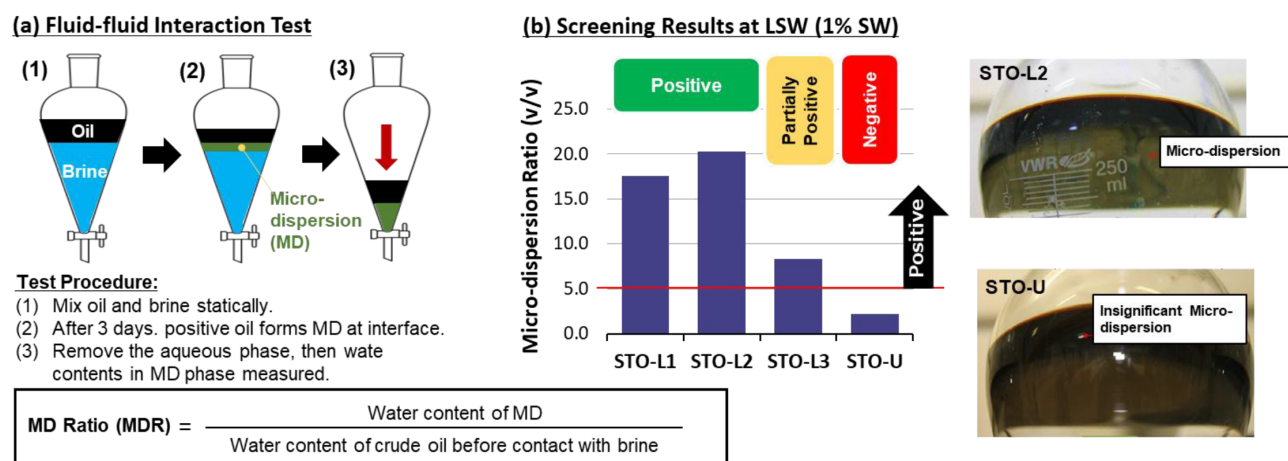


Figure 1—Fluid-fluid Interaction Test Procedure and Results: (a) Test Procedure; (b) Summary of FFI Test Results.

The previous FFI test screened the best positive oil, STO-L2, among the four candidates as shown in Figure 1(b) due to the highest MDR 20.3 v/v with a noticeable visual observation of MD formation at the interface. Therefore, STO-L2 was used as reference oil sample in this study.

Core Properties, Core Flood Apparatus, Procedure, and Way Forward from Results

Table 2 summarizes core plug characteristics. Three core plugs #1-3 taken from the preserved whole-core in the target reservoir were used to build a composite core. The core plugs were selected so that they could have similar poro-perm range to realize a homogeneous composite core condition. The length of composite core (i.e., number of core plugs) was decided to minimize capillary end effects. In addition, we assumed flow rates to ramp from 0.1 to 0.5 or 1.0 mL/min at the end of each water injection stage to reach the true residual oil saturation. This suppressed the potential capillary end effects within the system. Through the SEM/EDX analysis from the aspect of mineralogy, the core plugs #1 and #2 were concluded as Orthochemical limestone with calcite crystals in a microcrystalline cementing structure, while the core plug #3 showed small traces of magnesium (i.e., dolomite). While Tripathi et al (2023) reported that dolomite might reduce wettability alteration compared to calcite, the impact on composite core in our study was judged negligible due to consisting of almost calcite (i.e., 100% in cores #1-2 with trace level of dolomite in core #3). All the cores showed good permeability with maximum pore throat size of approximately 50 μm . Any invisible damage such as internal fractures and/or vugs were not observed by computed tomography (CT) scan analysis.

Table 2—Summary of Core Characteristics.

Core No.	#1	#2	#3
Porosity	0.184	0.204	0.189
Absolute Permeability (mD)	21.9	24.8	19.5
Length (cm)	8.51	6.66	8.47
Diameter (cm)	3.86	3.86	3.87
SEM Result	Porous	Porous	Porous
EDX Result	Pure calcite	Pure calcite	Traces of dolomite

In the previous studies (Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024b, 2024c), the LSW and/or hybrid LSW injection were conducted as a secondary mode and a tertiary mode after the secondary mode sea water injection using a typical core flood apparatus as shown in Figure 2(a). The hybrid LSW injectants were prepared to mix pure LSW (100 times diluted SW) and each additive such as DEK, DMK, and CuCl_2 . Core floods used the same set of composite core with restoration procedures (i.e., cleaned by solvent injection and then aged again) applied after each experiment to avoid core-dependent uncertainties. As a baseline injection rate, we assumed 0.1 ml/minute to reproduce practical injection velocity (1 ft/day) followed by bump flood rates up to 0.5 or 1.0 ml/min increased stepwise. The experimental temperature and pressure were set as reservoir temperature 230 °F (110 °C) and 500 psi higher than the re-livened oil saturation pressure. Back pressure and confining pressure were set as 2,000 and 3,000 psig, respectively. More details for a tertiary mode are as follows:

1. place core plugs in core holder with a Viton sleeve to make a composite core.
2. saturate core with formation water to stable differential pressure.
3. measure pore volume with lithium tracer at 100% Sw.
4. saturate core with dead crude oil to establish initial water saturation (just by pumping) and age for 4 weeks whilst monitoring dynamic differential pressure.
5. after 27 days of ageing process, displace dead oil with live oil (2 PV at a high rate and 1 PV at a low rate) on the 28th day.
6. inject SW at test rate (0.1 ml/minute), followed by bump flood up to 10 PV.
7. measure pore volume with lithium tracer at Sor (SW flood).
8. inject LSW or additive-doped LSW at test rate (0.1 ml/minute), followed by bump flood up to 10 PV.
9. measure pore volume with lithium tracer at Sor (LSW flood).
10. monitor differential pressure and fluid production constantly throughout experiment and calculate oil recovery factor.
11. after completed core flood run, inject toluene and methanol at 110 °C and 2,000 psi in successive manner over seven days as a cleaning process for re-use purpose.
12. displace methanol with 3% KCl prior to saturation process with formation water.

Figures 2(b) and (c) show core flood results for tertiary mode injection mode of pure LSW (CF01) and hybrid LSW (CF02). Finally, the oil recovery increments were correlated to MDR as shown in Figure 2(d). This finding was practically so useful because screening positive oil and/or designing additive condition can promise higher oil recovery. During detailed interpretation of oil recovery profiles, we were further motivated to investigate recovery mechanism and behavior because slow start-up of oil recovery was observed after switching tertiary mode in both CF01 and CF02.

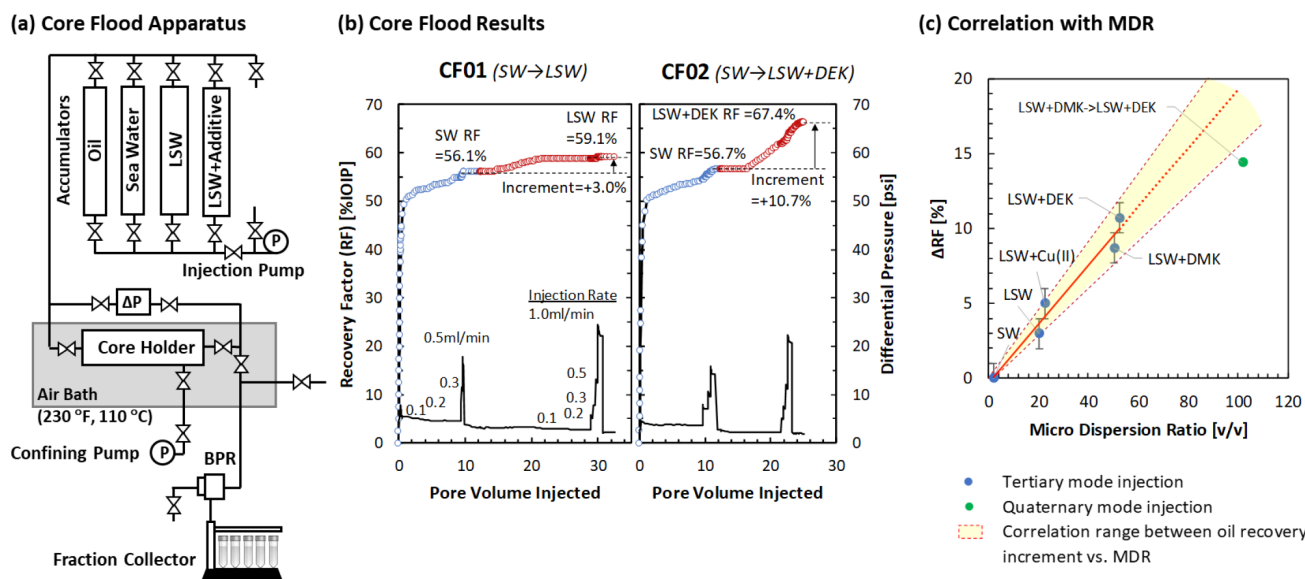


Figure 2—Core Flood Apparatus, Results, and Findings.

Microfluidic Chips Resembling Porous Media

Microfluidic flooding experiments were conducted to visualize displacement and investigate how the FFI mechanism enhances oil recovery. To replicate realistic reservoir conditions, a representative microfluidic chip was manufactured by reconstructing typical pore throat geometries based on a 2D CT-image stack scanned from the target carbonate core pieces. The chip surface was treated to reproduce carbonate-corresponding wettability characteristics. The chip design and manufacturing workflow is summarized in Figure 3. Micro-CT imaging was performed using a nanotom® (WayGate Technologies) with a resolution of 0.84-0.88 μm^3 .

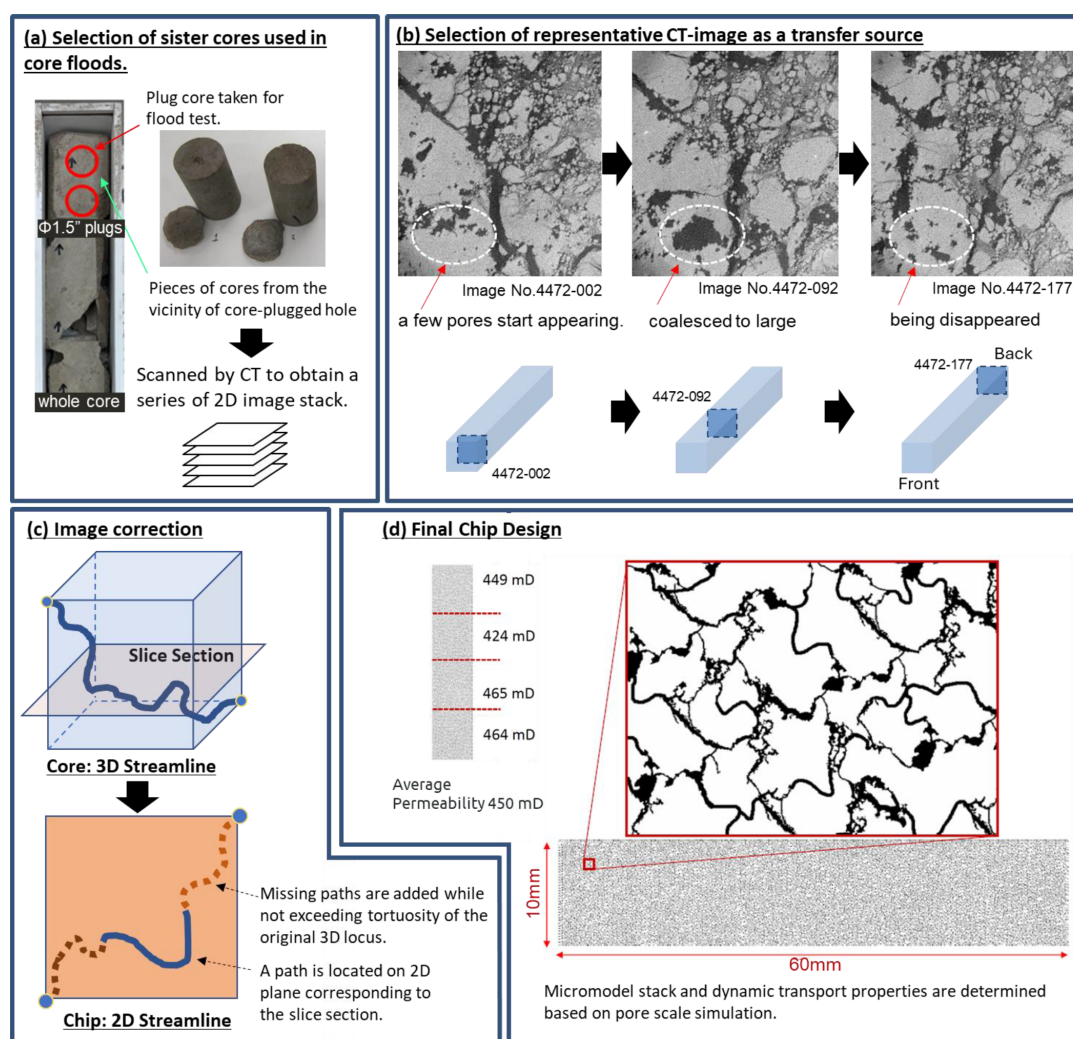


Figure 3—Workflow of Microfluidic Chip Designing and Manufacturing: (a) sister core selection; (b) representative CT-image as a transfer source; (c) image correction during conversion from three to two dimension; (d) final chip design.

Selection of sister cores. The reference pieces were collected from a vicinity of locations where core plugs were taken for core floods. This ensured that the microfluidic tests could use sister core information (see Figure 3(a)).

Selection of representative CT image. The core pieces were scanned using CT and all slice images were visually inspected to select the most representative one (see Figure 3(b)). Images with large holes, excess pore network connection, and/or too independent pore distribution were excluded.

Image correction. In general, when a three-dimensional object is converted to a two-dimensional image, a 2D slice section catches only a part of 3D information. As illustrated in Figure 3(c), a lost information of the path connecting two vertices in a cube model must be supplemented on the slice section. Otherwise, the connected streamline is converted to a non-connected one. To address this, missing connections were supplemented to preserve flow continuity and ensure that the 2D streamline tortuosity approximated the original value from 3D model as closely as possible.

Final chip design. As shown in Figure 3(d), a base pore throat design that was supplemented in the previous step is then replicated and extended to match the chip size (10 mm × 60 mm). Dynamic transport properties of the resulting micromodel were then estimated using pore scale simulation.

Microfluidics Flood Apparatus and Procedure

Microfluidic flooding experiments were performed to evaluate displacement behavior during LSW or hybrid LSW injection, in comparison to SW injection. The flooding apparatus is shown in Figure 4 (assembled by fluidXlab GmbH, maximum pressure: 10,000 psi, maximum temperature: 200 °C). The setup consists of a microfluidic chip, a chip holder equipped with a heating feature, absolute pressure sensors, a differential pressure sensor, high pressure microfluidic pumps, back pressure regulator (BPR), a high-speed camera and a microscope connected to a PC for continuous experimental and visual data recording. The BPR was controlled by nitrogen cylinder and set to a required experimental pressure.

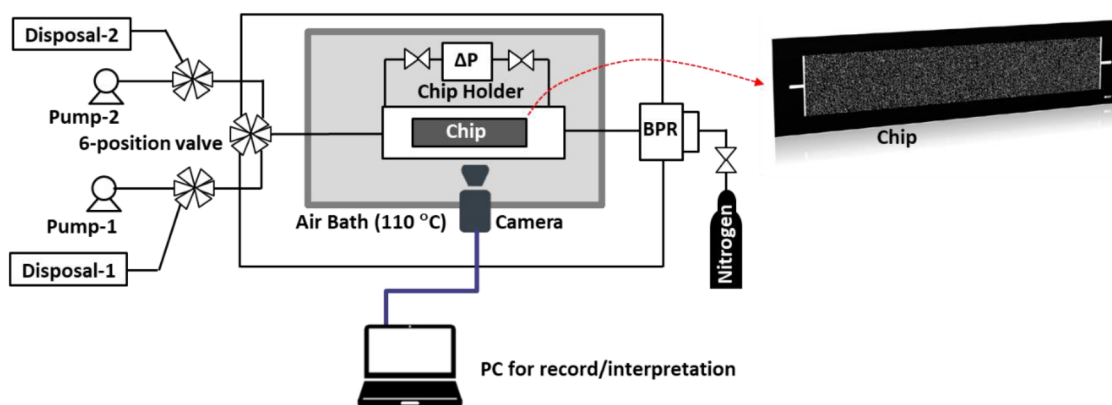


Figure 4—Microfluidics Flood Apparatus.

The flooding experiments were conducted in two phases: a series of preliminary tests, followed by a series of main tests. Preliminary tests used a ready-made micromodel designed to replicate a different reservoir. Tests were conducted at a temperature of 60 °C, same temperature as in the FFI tests that were conducted earlier. In contrast, the main tests were performed at reservoir temperature (i.e., 110 °C) and using a micromodel that was tailored for this study, with representative pore throat geometries derived from the target reservoir. The microfluidic chip used in a preliminary study featured larger pore space to facilitate visual observation of MD formation in the entire chip. The target-reservoir chip, however, was used for a sole purpose of capturing the realistic displacement behavior.

A summary of microfluidic flooding experiments is provided in Table 3. Initially, each chip was saturated to 100% by STO-L2 and then injections were performed stepwise at a flowrate corresponding to injection velocity of 1ft/day, with the exception of test MF04. In MF04, the chip was saturated by STO-L2 targeting initial water saturation of 38% to better represent realistic reservoir conditions. In the main tests, an ageing step was designed to allow sufficient reaction time for MD formation. Test MF05 assumed a 1-day ageing period, based on observation that most MD formation occurred during the first half of the 3-day FFI tests. However, this duration proved insufficient for complete reaction. Therefore, subsequent tests MF06 and MF07 included a 1-week ageing period to ensure a complete reaction time with safety margin.

Table 3—Summary of microfluidic floods.

Flood No.	Injectant Sequence	Temperature (°C)	Type of Chip
MF01	① SW	60	larger pore-replicated chip
MF04	① LSW	60	larger pore-replicated chip
MF05	① LSW→② Ageing (1 day)→③ LSW	60	larger pore-replicated chip
MF06	① SW→② LSW+DEK→③ Ageing (1 week)→④ LSW+DEK	110	target-carbonate structure chip
MF07	① SW→② LSW+DEK→③ Ageing (1 week)→④ LSW+DEK (2PV)→⑤ SW	110	target-carbonate structure chip

Microfluidics Flood Results

Reference Core Piece

Three sets of core pieces were collected from a vicinity of each core plug #1-3 sampling sites, respectively. These pieces were scanned using micro-CT to obtain 2D slice images, as shown in Figure 5. The characteristics of the core plugs and their corresponding pieces are summarized in Table 4.

The 2D images revealed distinct differences among the core pieces. Piece-1 exhibited visible pore spaces and/or connected pore networks. Piece-2 appeared dense and tight. Piece-3 showed isolated pores without connected networks. Consistent with the observations, Piece-1 showed the smallest equivalent diameter in its 3D model among the three pieces. The dense appearance of Piece-2 was confirmed again by the smallest range porosity profile in the 3D model. According to comparison of the 3D porosity profiles, a certain volume of isolated pore exists in the Piece-2 and Piece-3, whereas Piece-1 contained negligible level of isolated space. Additionally, Piece-1 showed significant heterogeneity in porosity across different locations, contributing to its high hydraulic tortuosity. Based on these observations, Piece-1 was selected as a reference for manufacturing a microfluidic chip. However, due to its heterogeneity, special case was taken to select a representative 2D slice that accurately reflects the 3D pore structure.

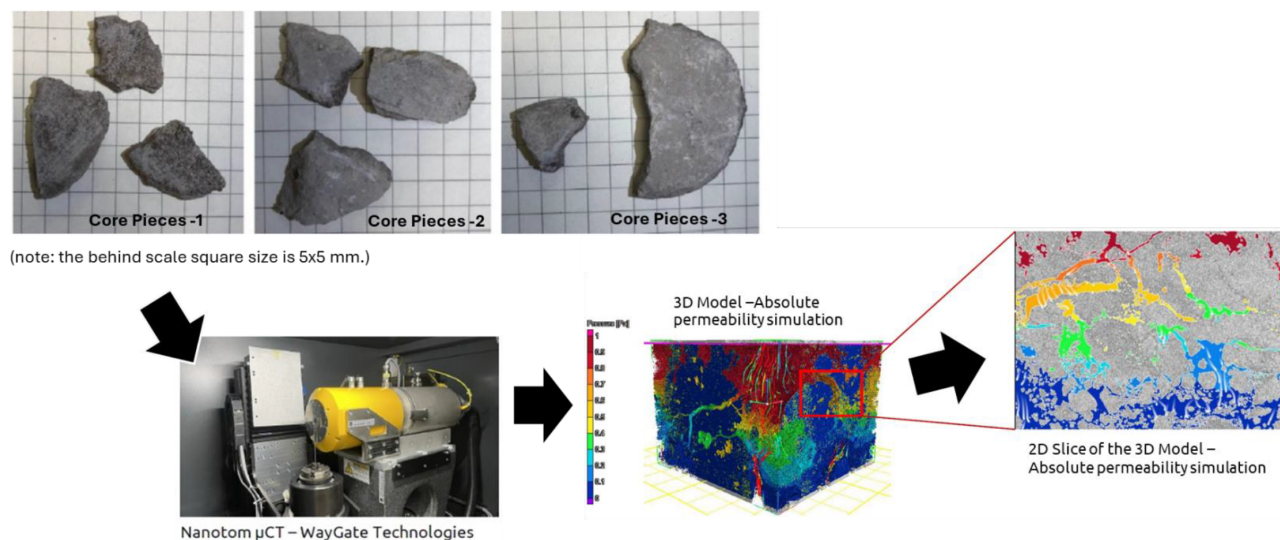
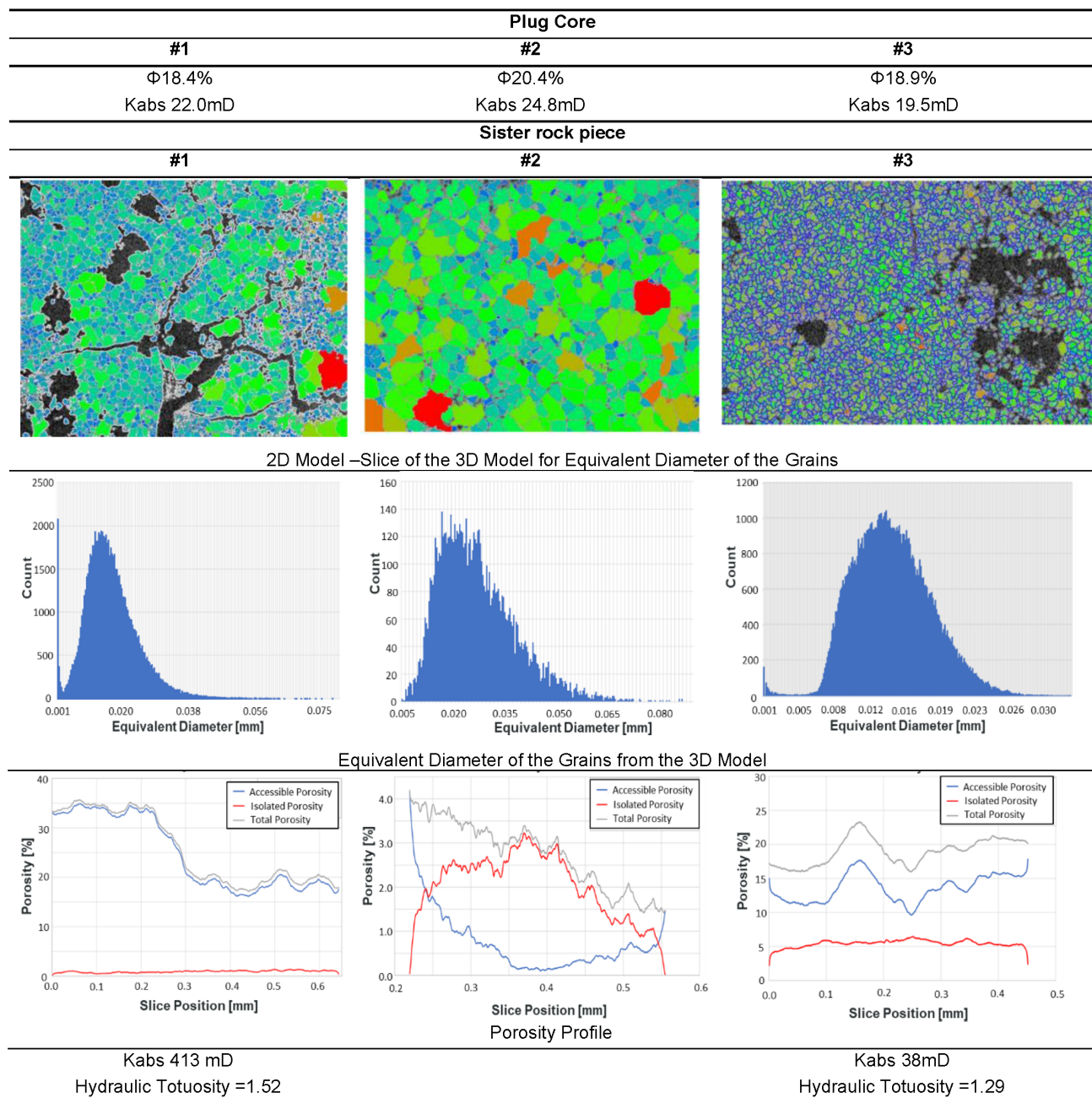


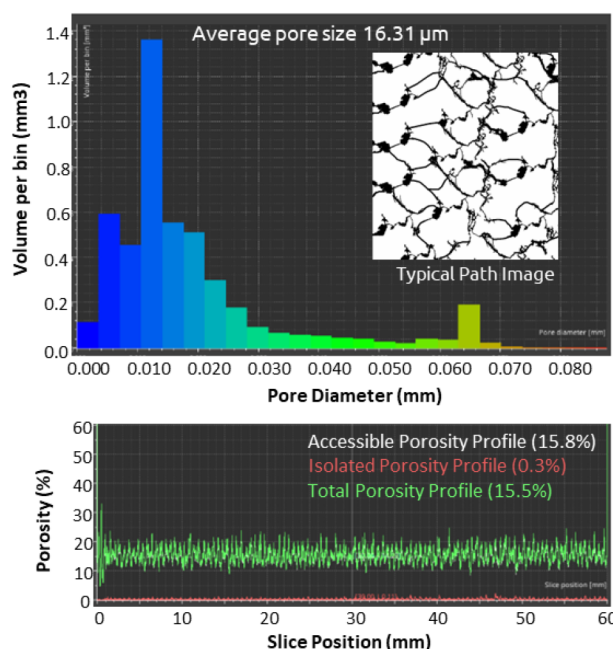
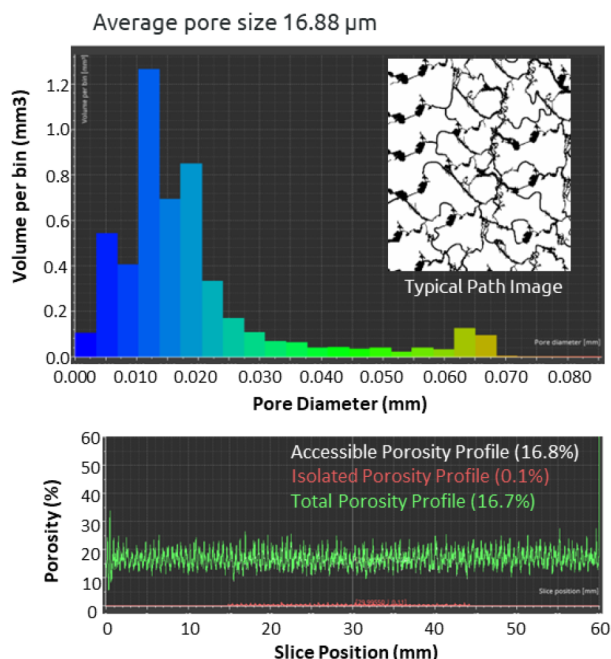
Figure 5—Sequence of Core Information Collection for Microfluidic Chip Design: (1) sister core pieces collected from whole core; (2) micro CT-image scan; (3) 3D model image; (4) 2D slice image from the 3D model.

Table 4—Summary of Piece Characteristics.



Reservoir Chip

Based on the representative 2D slice information from the core Piece-1, the chip design was finalized through a trial-and-error process, as shown in Figure 6. The initial design (Design #1) supplemented missing streamlines with relatively straight paths. This made simple streamlines but resulted in a lower tortuosity of 1.17, compared to the reference tortuosity of 1.52 from the 3D model. To improve this, a second design (Design #2) was created by supplementing the missing streamlines with curved paths. This resulted in increased tortuosity of 1.29, which was closer to the 3D reference value. The supplemented lines were guided by the shapes of existing pore paths. Although a third design was considered with more exaggerated curvature and longer paths, it was ultimately rejected due to its excessively artificial appearance which did not reflect the natural geometry observed in the core plug.

(a) Properties of Chip Design #1**(b) Properties of Chip Design #2****(c) Properties of Core and Artificial Chip Structure**

Property	Plug Core No.1	Chip Design #1	Chip Design #2
Accessible Porosity [%]	24.72	15.52	16.74
Isolated Porosity [%]	0.94	0.25	0.09
Total Porosity [%]	25.66	15.77	16.83
Mean Pore Size [μm]	13.22	16.31	16.88
Absolute Permeability [mD]	412	447	450
Tortuosity [-]	1.52	1.17	1.29

Figure 6—Properties of Reference Core and Artificial Chip Structure: (a) properties of chip design #1; (b) properties of chip design #2; (c) comparison between plug core and chip designs.

Therefore, Design #2 was selected for use in this study.

Preliminary Microfluidic Flood Results (MF01 and MF04)

Microfluidic flooding experiments were performed in two stages. As a preliminary evaluation, experiments MF01 and MF02 were conducted to distinguish pure LSW effect compared with SW. The flood results are shown in Figure 6(a)(b). A notable difference between the two runs was observed in differential pressure behaviors. The differential pressure in MF01 remained low and stable around 10 psi. In contrast, MF04 exhibited a significant increase in differential pressure, exceeding 60 psi. Visual observation of magnified pore images revealed the presence of emulsions in MF04, indicated by yellow arrows, whereas no emulsions were observed in MF01. These findings support the hypothesis that LSW EOR mechanism can lead to better oil sweep efficiency due to mobility improvements caused by emulsion induced blocks. The block of preferential flowing paths led to increased differential pressure. The observation of emulsion formation in MF04 was also consistent with MD formation in earlier FFI tests. Another hypothesis, discussed by Kar et al. (2022), involves the formation of viscoelastic materials around the interface of oil and brine. This may also contribute to mobility control improvement.

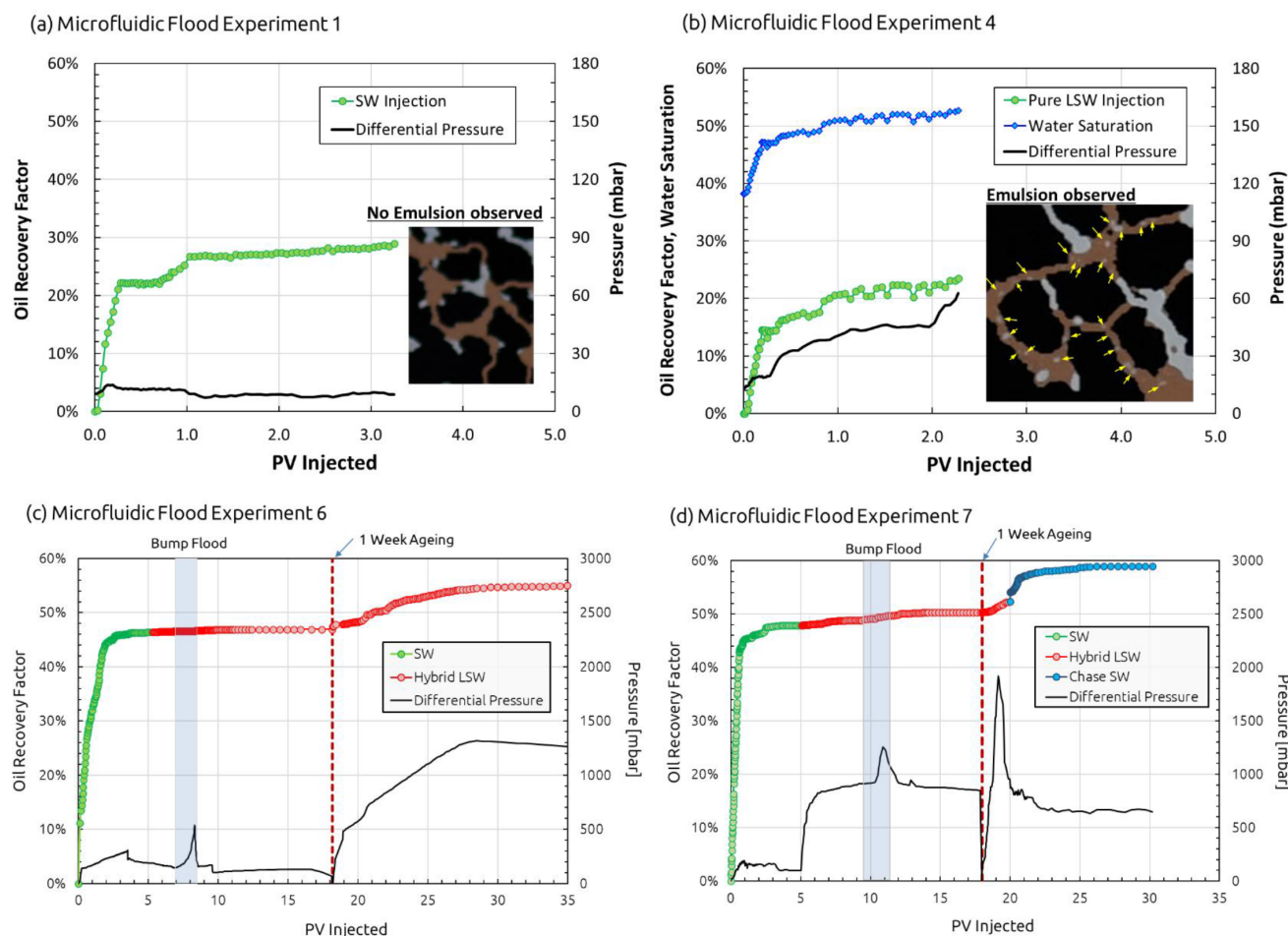


Figure 7—Microfluidic Flood Results: (a) MF01, secondary mode injection of SW; (b) MF04, secondary mode injection of pure LSW with 1-day ageing; (c) MF06, tertiary mode injection of hybrid LSW with 1-week ageing; (d) MF07, tertiary mode injection of hybrid LSW with 1-week ageing followed by chase SW.

Main Microfluidic Flood Results (MF06 and MF07)

As part of the main evaluation, MF06 and MF07 were conducted to investigate the hybrid LSW injection mechanism. The microfluidic flooding results are shown in Figure 6(c)(d).

During the secondary mode injection of SW, oil recovery reached approximately 46.4 % in MF06 and 47.9% in MF07, as both tests used the same microfluidic chip. Similar trends were observed in both differential pressure and oil recovery profiles before and after the ageing stage. A small increase in oil recovery was seen during tertiary hybrid LSW injection prior to ageing. However, a further notable oil production occurred after ageing, accompanied with a clear differential pressure increase. In MF06, tertiary hybrid LSW injection led to an additional +8.4% of oil recovery, with the pressure differential rising to 9 times the pre-ageing value. Similarly, in MF07, tertiary mode injection of hybrid LSW followed by a chase SW injection resulted in an oil recovery gain of +8.6% after the ageing. The differential pressure after the ageing reached 9 times the level of secondary SW injection and 2 times that of the first half of tertiary hybrid LSW injection. As in the preliminary test, the formation of emulsion blockages and/or viscoelastic interfacial structures were considered as possible mechanisms contributing to improved mobility control.

When compared with the previous core flood results (i.e., oil production commenced only after injecting 2-4 PV tertiary mode LSW or hybrid LSW injection in CF01 and CF02) in Figure 2(b), the most important finding was to commence oil production earlier, within just 1-2 PV, after the ageing in MF06 and MF07. This highlights the importance of considering the reaction times when designing flood.

In a more detailed further analysis, we examined magnified images of chip at different stages of injection. [Figure 8](#) shows the injection front behaviors during secondary mode SW injection in MF07. Since the chip was placed horizontally, gravitational segregation did not influence the flow pattern. The flood started from 100% oil saturated condition. The first image at 0.04 PV shows small water encroachment (i.e., pale blue color) from left-side (inlet). By 0.15 PV (second image), a dominant water invasion path began to form, and by 0.4 PV (third image) a clear fingering phenomenon was visible. At 0.5 PV, water reached the outlet-side (right-side), and the water-flooded region progressively expanded across the chip. The final image at 2.8 PV shows the entire chip appearing uniformly water-flooded. The smallest baseline pattern of paths which mimics real carbonate heterogeneity likely caused fingering locally. However, the overall chip repeatedly extended with copying the original pattern was considered to make apparent homogeneous flood front at the macro scale.

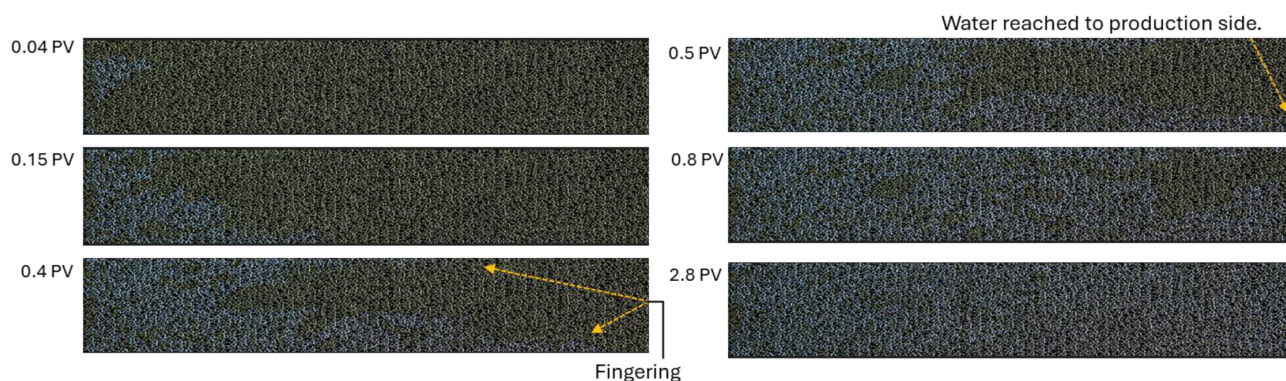


Figure 8—Images during secondary sea water injection stage in microfluidic flood #07.

Images of oil and hybrid LSW in contact during the ageing stage in MF07 are shown in [Figure 9](#). While no significant changes were observed in the Days 1-2, emulsions became visible on Day 3. This observation is consistent with the findings from the FFI tests, which confirmed MD formation after 3 days of contact.

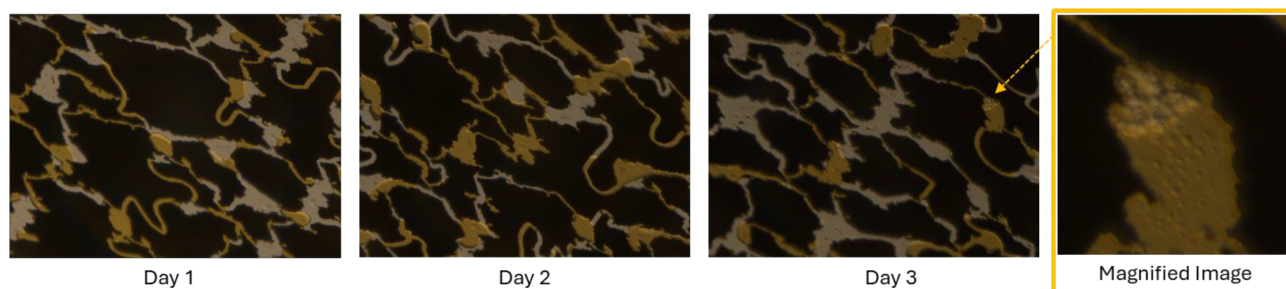


Figure 9—Images during ageing stage in microfluidic flood #07.

The flow behavior of oil droplets in porous media were captured at the chase SW injection stage in MF07, as shown in [Figure 10](#). Oil droplets located in larger pores were observed moving through narrow paths towards outlet-side.

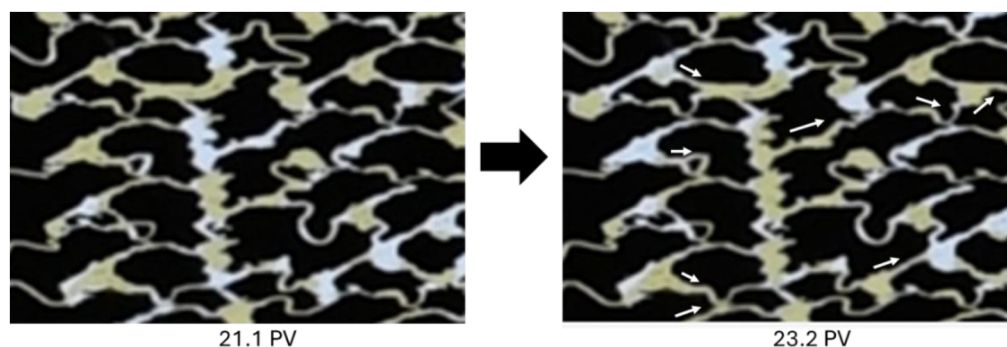


Figure 10—Images during chase SW injection stage in microfluidic flood #07.

Way Forward

Through our previous studies (Yonebayashi et al. 2024a, 2024b), we have tried evaluating viscosity variation for various fluids such as original oils and MD fluids collected from FFI tests. Although we have not reached a conclusion yet by the past studies, the differential pressure behaviors during the microfluidic flood tests in this study imply varying viscosity and/or interfacial viscoelasticity of LSW/hybrid LSW contacted oils. In the latest research paying attention on crude oil-water interfacial viscoelasticity as a more dominant effect on oil recovery compared with a general EOR mechanism governed by wettability alteration, Chávez-Miyauchi et al. (2025) reported an advantage of synthetic tuning elasticity at the interface of crude oil and brine. In their work, small amount of polymetric surfactant was used to accelerate improving the viscoelasticity significantly while a simple contact of crude oils and brines showed insignificant results. Here, there are still debatable points. A key interfacial reaction of MD formation highly depends on characteristics of crude oils in a series of previous LSW EOR studies (Masalmeh et al. 2019; Alhammadi et al. 2021; Hiraiwa et al. 2023; Yonebayashi et al. 2024a, 2024b). Therefore, a way forward will be an experimental investigation of viscosity/viscoelasticity change using positive oils that will be screened previously.

The previous research of key compositions in positive oils for MD-forming mechanism identified polar heavy ends such as polyaromatic asphaltenes and resins that work as surface-active molecules (Yonebayashi et al. 2024d, 2025). Those surface-active materials were considered to form micro-dispersion phase by creating rigid viscoelastic films on the oil/water interface. Yarranton et al. (2007) reported that elasticity of asphaltene-formed film increased with increasing ageing time. This supports the findings from our microfluidic flood experiments. In addition, Yarranton et al. (2007) found the elasticity increased with increasing other factors such as asphaltene bulk concentration. Thus, the way forward will be performed using several samples containing various heavy-end concentrations from the perspective of geochemical comparison.

Conclusion

To understand pure/hybrid low salinity water (LSW) EOR mechanism, we conducted a total of four microfluidic flood experiments using a positive oil that was screened as LSW-reactive oil during fluid-fluid interaction (FFI) test. LSW was prepared by diluting sea water (SW) to 1%. A microfluidic chip was manufactured by reproducing a typical pore throat/network pattern selected from a sister carbonate core that was used in the past core flood tests. The experimental results and findings are summarized as below:

- A representative reservoir chip was reproduced by selecting the most typical pore throat/network pattern. The selection was performed based on micro-CT scanned images so that non-network/independent pores, too tight/dense rock, and/or excess large vugs/pores could be avoided. The original 3D core tortuosity information was closely maintained in the 2D chip tortuosity by applying appropriate corrections.

- Using a ready-made shaped chip that had larger space pores than that in the target carbonate reservoir-replicated chip, preliminary tests were conducted evaluating secondary mode injection of SW vs LSW at 60 °C (FFI test temperature). The visual observation captured emulsions in oil phase during LSW injection stage. The emulsion formation was consistent with the previous findings of micro-dispersion (MD) formation in FFI tests. The differential pressure records maintained stably at low level (10 psi approx.) in the SW injection case while noticeably increased over 60 psi in the LSW injection case.
- Using a replicated chip representing the target reservoir, main microfluidic tests were conducted to evaluate tertiary mode injection of hybrid LSW (containing 2wt% diethyl ketone) at the reservoir temperature of 110 °C. To secure MD-forming reaction time, 1-week ageing period was included in the procedure of hybrid LSW injection stage. After the ageing, a baseline run continued hybrid LSW injection to the end of flood while another run switched to chase SW injection at the last stage. Both runs achieved additional oil recovery increment (+8.4-8.6 % IOIP) after the ageing, therefore, securing reaction time was a crucial setting in the small experimental scale. The differential pressure profiles showed a clear increasing trend by hybrid LSW injection.
- The microfluidic floods under appropriate experimental settings (i.e., sufficient reaction time) commenced to produce additional oil production quicker than the lead injection period in core floods under the continuous injection process that was shorter than the necessary reaction period.
- The emulsion observation with differential pressure increasing trend implied varying viscosity and/or forming viscoelastic materials at the interface of oil and pure/hybrid LSW in contact. A way forward will be further investigation of these phenomena in LSW EOR mechanism.

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